

NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Mechanical Engineering Department

Time: 2 hrs

Marks: 30

Mid semester exam (May 2023)

Elements of Mechanical Engineering (ME22101)

Assume missing data if any. Answer all questions.

1. Describe nuclear energy and wind energy. [10]
2. Explain in detail using suitable diagram about Babcock Wilcox boiler. [10]
3. Differentiate between renewable resource and non-renewable resource. [5]
4. Write a short note using suitable diagram about superheater of boilers. [5]

GYB

NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science and Engineering
MID SEMESTER EXAMINATION, Jan-Jul 2023

B.Tech.: ME

Course Name: Fundamentals of Information Technology

DOE: 15/05/2023

Maximum Time: 2 hours

Course Code: CS22101

Max. Marks: 30

Instruction:

1. Attempt all questions.
2. Assume any suitable data, if necessary.
3. The Marks, CO (Course Outcome) and BL (Bloom's Level) related to questions are mentioned on the right-hand side margin.

S.N.	Questions	Marks	CO	BL
1.	Define computer. Explain any five characteristics of a digital computer.	[4]	CO1	Remember
2.	What are the factors to consider when categorizing computer? Explain categories of Computer.	[1+3=4]	CO1	Understand
3.	What do you mean by primary and secondary memory? Explain types of RAM in computer systems.	[1+3=4]	CO1	Understand
4.	What are the two key factors that characterize the memory? Explain with a neat diagram.	[1+3=4]	CO2	Remember
5.	Explain the working of a magnetic disk. Define the terms <i>seek time</i> , <i>latency time</i> , <i>data transfer rate</i> and <i>access time</i> .	[2+2=4]	CO2	Understand
6.	a) Convert $(159)_{10}$ to the octal number system. b) Convert $(2C4)_{16}$ to the decimal number system <i>e708</i> c) Convert $(546)_8$ to the hexadecimal number system <i>166</i>	[2+2+2=6]	CO2	Understand
7.	Answers the following questions: a) "Two processors are compatible". How do you deduce this statement? b) "The performance of RAM is affected by the access speed and the data transfer unit size". Explain.	[2+2=4]	CO2 CO2	Analyze Analyze

All the best



NATIONAL INSTITUTE OF TECHNOLOGY PATNA

Department of Mechanical Engineering

Workshop Practice-II (ME22105)

Mid-Semester Examination (January- June 2023)

Date: 11-05-2023

Timing: 02:00 PM - 04:00 PM Session: 2022-23, B. Tech (2nd Sem.) Max. Marks: 30

Instructions:

- Attempt any **Four** out of **Five** questions.
- The marks related to the questions are mentioned on the right-hand side margin.
- The CO and BL associated with the questions are also mentioned.

	Questions	Marks	CO	BL
1.	Describe the recrystallization process and recrystallization temperature. What are the differences between hot working and a cold working process?	7.5	2	1
2.	Explain the various smithy forging operations. Describe the defects in the forging process.	7.5	2	1
3.	Explain the principle of extrusion and various types of extrusion processes.	7.5	2	1
4.	Explain the mechanics of the rolling process. Describe the roll separating force and power in the rolling process.	7.5	2	2
5.	In a single pass rolling operation, a 20 mm thick plate with plate width of 100 mm, is reduced to 18 mm. The roller radius is 250 mm and rotational speed is 10 rpm. The average flow stress for the plate material is 300 MPa. What is the power required for the rolling operation in kW ?	7.5	2	2

Best wishes



NATIONAL INSTITUTE OF TECHNOLOGY PATNA
(An Institute under Ministry of HRD, Govt. of India)
Mid Semester Examination- 9th May 2023

Program: B.Tech 1st Year, ME

Course Code : CH22101

Course name: Engineering Chemistry

Time: 2.00 Hours

Full Marks: 30

Attempt All Questions

- 1) Derive the following expression of ΔH and ΔS

(5) [CO2]

$$\Delta H = -nF[E - T\left(\frac{\partial E}{\partial T}\right)_P]$$

$$\Delta S = nF\left(\frac{\partial E}{\partial T}\right)_P$$

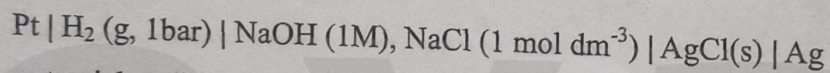
What are the unit of ΔG , ΔH , and ΔS if F is expressed in coulomb mol^{-1} and E in volt?

- 2) Explain the essential feature on which the potentiometric titration is based. Describe them with suitable example.

(3)[CO2]

- 3) For the cell

(4) [CO2]



The potential at different temperatures are

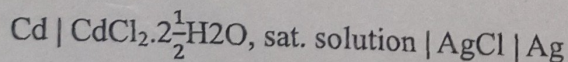
$t/^{\circ}\text{C}$	20	25	30
E/V	1.047 68	1.051 88	1.056 08

- a) Calculate the value of pK_w^0 at these temperatures.
b) Calculate the enthalpy and entropy of ionization of water at 298 K.

Given: $E^0_{\text{Cl}^- | \text{AgCl} | \text{Ag}} = 0.2225 \text{ V}$

- 4) The emf of the cell

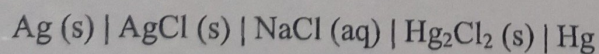
(4) [CO2]



At 298 K is 0.675 33 V and $(\partial E / \partial T)_P$ is $-6.5 \times 10^{-4} \text{ V K}^{-1}$. Calculate ΔG , ΔH and ΔS for the cell reaction at 298 K.

- 5) Consider the following cell arrangement.

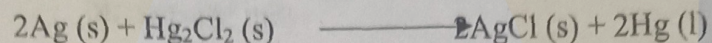
(4) [CO2]



- a) Write down the half-cell reaction.
b) The standard emf of the cell at various temperatures are as follows:

t/K	291	298	303	311
E ⁰ /mV	43.0	45.4	47.1	50.1

Calculate ΔG , ΔH and ΔS for the cell reaction



- 6) The limiting molar conductivities of NaI, NaCH₃CO₂ and Mg(CH₃CO₂)₂ are 12.69 mS m² mol⁻¹, 9.10 mS m² mol⁻¹, and 18.78 mS m² mol⁻¹, respectively (all at 25°C). What is limiting molar conductivity of MgI₂ at this temperature. (4) [CO2]
- 7) Discuss the Kohlrausch's law and Ostwald's dilution law. (3) [CO2]
- 8) Derive $v_{\pm}\lambda_{\pm} = t_{\pm}^0\Lambda_m^0$ (3) [CO2]

where, $v_{\pm}$ = cation and anion charge

$t_{\pm}$ = transport number

$\lambda_{\pm}$ = molar conductivity of cation and anion

Λ_m^0 = limiting molar conductivity.

National Institute of Technology Patna
Department of Mathematics

Mid Semester Examination : May 2023

MA22101 : Engineering Mathematics

Branch: B.Tech(ME-Group B) & DD(ME) (2nd-Semester)

Maximum Marks: 30

Time: 02.00 hours

Questions 1 & 2 are compulsory · Answer any five out of the remaining questions

1. For what values of $a \in \mathbb{R}$ and $b \in \mathbb{R}$, the following system of equations in unknowns x, y and z , has (i) no solution, (ii) a unique solution, and (iii) infinitely many solutions:

$$ax + by + 2z = 1, \quad ax + (2b - 1)y + 3z = 1, \quad ax + by + (b + 3)z = 2b - 1?$$

Also, find the solutions whenever they exist.

[5 Marks]

2. Find eigenvalues and corresponding eigenvectors for all real values of k for the matrix $B =$

$$\begin{bmatrix} 1 & 0 & k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and also find values of k for which the matrix B is diagonalizable.

[5 Marks]

3. Find the inverse of the matrix using Gauss-Jordan method:

$$\begin{bmatrix} 1 & -1 & 0 & 0 \\ 2 & 4 & 0 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & -6 & -2 \end{bmatrix}$$

[4 Marks]

4. Find the basis and dimension of the null space of the following matrix:

$$A = \begin{bmatrix} 1 & -1 & 0 & 1 & -2 \\ 1 & 4 & 3 & -1 & -4 \\ 2 & 3 & -4 & -7 & -3 \\ 2 & -2 & 0 & 2 & -4 \end{bmatrix}$$

[4 Marks]

5. Determine if the set $T = \{1, x^2 - x + 5, 4x^3 - x^2 + 5x, 3x + 2\}$ spans the vector space of polynomials with degree 4 or less.

[4 Marks]

6. Let $t \in \mathbb{R}$. Discuss the linear independence of the following three vectors over $\mathbb{R}^3$: $u = [1, 1, 0]^t$, $v = [1, 3, -1]^t$ and $w = [5, 3, t]^t$.

[4 Marks]

7. Find the orthogonal projection of v onto the subspace W spanned by the vectors u_1, u_2 and u_3 , where $v = [3, -2, 4, -3]^t$, $u_1 = [1, 1, 0, 0]^t$, $u_2 = [1, -1, -1, 1]^t$ and $u_3 = [0, 0, 1, 1]^t$. Additionally, determine the distance between v and the subspace W .

[4 Marks]

8. Find a homogeneous system whose solution space is exactly spanned by the vectors $[1, 1, 2, 1, 1]^t$, $[1, 2, 1, 4, 3]^t$ and $[3, 5, 4, 9, 7]^t$.

[4 Marks]

***** ALL THE BEST *****